# Fiscal and Monetary Regimes

## The story of determinacy and necessary conditions

based on Leeper (1991) and Blanchard & Kahn (1981)

Warsaw Reading Group, October 2026

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# Motivation

- We already have introduction to New Keynesian models
- What we have is a three equation NK model: with Euler equation, with NK Phillips curve and with certain policy rule for setting interest rates (which we called Taylor rule)
- For my part, I want to add another element: government which owns bonds, collects taxes, spends money and has another policy rule
- The target will be to show what results from 2 policy rules and how they collectively choose the equilibrium (or do they?)

# Economy with government

We will begin with deterministic economy, in which $y_t = y$. There are two assets: real balances $m_t$ and government bonds, which yields nominal yield of $R_t$. The consumer chooses the vector $(c_t, m_t, b_t)$ to maximise:

$$\sum_{t=0}^{\infty} \beta^t [\log(c_t) + \log(m_t)]$$

under the budget constraint:

$$c_t + m_t + b_t + \tau_t = y_t + R_{t-1} b_{t-1} + \frac{M_{t-1}}{p_t}$$

The markets clear for $y_t = c_t + g_t$

# First order conditions

The emerging FOCs may be reduced to Fisher equation:

$$\frac{1}{R_t} = \beta \mathbf{E}_t(\frac{1}{\pi_{t+1}})$$

and money-demand relationship:

$$m_t = c(\frac{R_t}{R_t - 1})$$

as well as transversality condition for real government debt (which just says that the value of government debt is equal to the present value of futures surpluses), which results in:

$$b_t = \sum_{s=0}^{\infty}(\prod_{j=0}^{s} \pi_{t+j+1} R_{t+j}^{-1})(\tau_{t+s-1} - g_{t+s-1} + \frac{M_{t+s+1} - M_{t+s}}{p_{t+s+1}})$$

# Policy rules

We may now add the monetary authority setting interest rates $R_t$ under rule:

$$R_t = \alpha_0 + \alpha\pi_t + \theta_t$$

and government setting taxes under rule:

$$\tau_t = \gamma_0 + \gamma b_{t-1} + \psi_t$$

With both $\theta_t$ and $\psi_t$ being AR(1) processes.

# Log-linearized model

Following log-linearization of this problem around the steady-state solution, we get the set of equations:

$$\mathbf{E}_t(\tilde{\pi}_{t+1}) = \alpha\beta\tilde{\pi}_t + \beta\theta_t$$

We add this to government balance motion to get:

$$\phi_1\tilde{\pi}_t + \tilde{b}_t + \phi_2\tilde{\pi}_{t-1} - (\beta^{-1} - \gamma)\tilde{b}_{t-1} + \phi_3\theta_t + \psi_t + \phi_4\theta_{t-1} = 0$$

So all in all, we can write it in matrix form:

$$\begin{bmatrix}\mathbf{E}_t(\tilde{\pi}_{t+1})\\ \mathbf{E}_t(\tilde{b}_{t+1})\end{bmatrix} = \begin{bmatrix}\alpha\beta & 0\\ \kappa_\pi & \beta^{-1} - \gamma\end{bmatrix}\begin{bmatrix}\mathbf{E}_t(\tilde{\pi}_t)\\ \mathbf{E}_t(\tilde{b}_t)\end{bmatrix} + \begin{bmatrix}\theta'_t\\ \psi'_t\end{bmatrix}$$

# Little algebra

Let's make a short break for the equation of the form:

$$\begin{bmatrix} \mathbf{E}_t(x_{t+1}) \\ \mathbf{E}_t(y_{t+1}) \end{bmatrix} = A \begin{bmatrix} \mathbf{E}_t(x_t) \\ \mathbf{E}_t(y_t) \end{bmatrix} + B \begin{bmatrix} \varepsilon_t^x \\ \varepsilon_t^y \end{bmatrix}$$

, in which $x_t$ denotes jump variables, while $y_t$ denotes state variables. What decides the stability of such an equation? Well, we can diagonalize $A = PDP^{-1}$ and get

$$P^{-1} \begin{bmatrix} \mathbf{E}_t(x_{t+1}) \\ \mathbf{E}_t(y_{t+1}) \end{bmatrix} = DP^{-1} \begin{bmatrix} \mathbf{E}_t(x_t) \\ \mathbf{E}_t(y_t) \end{bmatrix} + P^{-1}B \begin{bmatrix} \varepsilon_t^x \\ \varepsilon_t^y \end{bmatrix}$$

So for each variable, we have an eigenvalue.

# Blanchard-Kahn conditions

Now let us decide, why are paths unique? Well, for one: they should go to the steady state (after all, we think of steady state equilibrium), so they shouldn't explode. However, we also would like to have a uniqueness — namely, that one has to choose the same path in order to get to the same point.

The answer? Saddle path — namely, the matrix of evolution must be defined so that control variables explode, while state variables remain bounded. We may check by checking the length of eigenvalues. The number of eigenvalues greater than 1 must be equal to the number of control variables, while the number of eigenvalues inside the unit circle should be equal to the number of state variables.

# Existing regimes

Ok, so looking back at the matrix, we can see that there are 2 eigenvalues and therefore 4 possible combinations:

- $\alpha\beta > 1$ and $\beta^{-1} - \gamma > 1$ (active fiscal and active monetary)
- $\alpha\beta < 1$ and $\beta^{-1} - \gamma > 1$ (active fiscal and passive monetary)
- $\alpha\beta > 1$ and $\beta^{-1} - \gamma < 1$ (passive fiscal and active monetary)
- $\alpha\beta < 1$ and $\beta^{-1} - \gamma < 1$ (passive fiscal and passive monetary)

As we discussed, the first one will be explosive and the fourth one will not yield unique solution. Therefore, we focus on the second and third.

# Characteristics of regimes

The regime PF and AM is the classical one: the government controls debt and therefore, the inflation is solved forward without any consideration for it. The monetary policy does not depend on the fiscal shocks. However, this does not mean that there is no fiscal authority! The solution is possible only because the fiscal policy takes care of debt by adjusting taxation.

The second regime is more complicated. With AF and PM the lack of relationship between inflation and fiscal shocks disappear. Both fiscal shocks and monetary shocks may also affect real variables, as well as present and future inflation. Monetary shocks may lead to both lower and higher inflation, depending on the other variables!

# Conclusion

- For the debt kept in control, the NK model (with CB aggressive enough) is sufficient — CB may act according to some rule, and the equilibrium will be unique.
- However the ability to stabilize inflation by CB only is lost under the exogenous deficits without reaction of government — see Sargent, Wallace (1981)
- The price level is therefore defined either by monetary authority (classical models with monetary authority) or by fiscal authority (FTPL — Fiscal Theory of Price Level)
- The difference in source of determinacy comes from analyzing Blanchard-Kahn conditions for the equation.